RECURRENCE RELATION

- A recurrence is an equation or inequality that describes a function in terms of its values on smaller inputs.
- To solve a Recurrence Relation means to obtain a function defined on the natural numbers that satisfy the recurrence.

General forms

$T(n) = aT(n/b) + f(n)$

$T(n) = aT(n-b) + f(n)$

- For Example, the Worst Case Running Time T (n) of the MERGE SORT Procedures is described by the recurrence.

$T(n) = \theta(1)$ if $n=1$

$2T(n/2) + \theta(n)$ if $n>1$

There are four methods for solving Recurrence:

1. Iteration Method
2. Substitution Method
3. Recursion Tree Method
4. Master Method

ITERATION METHOD

- It means to expand the recurrence and express it as a summation of terms of n and initial condition.

Example1: Consider the Recurrence

T (n) = 1 if n=1

= 2T (n-1) if n>1

Solution:

T (n) = 2T (n-1)

= 2[2T (n-2)] = 2^2T (n-2)

= 4[2T (n-3)] = 2^3T (n-3)

= 8[2T (n-4)] = 2^4T (n-4) (Eq.1)

Repeat the procedure for i times

T (n) = 2^i T (n-i)

Put n-i=1 or i= n-1 in (Eq.1)

T (n) = 2^{n-1} T (1)

= 2^{n-1} .1 {T (1) =1 .....given}

= 2^{n-1}

Example2: Consider the Recurrence

T (n) = T (n-1) +1 and T (1) = θ (1).

Solution:

T (n) = T (n-1) +1

= (T (n-2) +1) +1 = (T (n-3) +1) +1+1

= T (n-4) +4 = T (n-5) +1+4

= T (n-5) +5= T (n-k) + k

Put n-k=1

K=n-1

Where k = n-1

T (n-k) = T(n-(n-1))=T (1) = θ (1)

T (n) = θ (1) + (n-1) = 1+n-1=n= θ (n).

SUBSTITUTION METHOD:

The Substitution Method Consists of two main steps:

1. Guess the Solution.
2. Use the mathematical induction to find the boundary condition and shows that the guess is correct.

Example1

Solve the equation by Substitution Method.

$$T(n) = T(n/2) + n$$

We have to show that it is asymptotically bound by O (log n).

Solution:

For T (n) = O (log n)

We have to show that for some constant c

T (n) ≤c logn.

Put this in given Recurrence Equation.

$$T(n) \le c \log(n/2) + 1$$

$$\le c \log(n/2) + 1 = c \log n - c \log_2 2 + 1$$

$$\le c \log n \text{ for } c \ge 1$$

Thus T (n) =O logn.

Example2

Consider the Recurrence

T (n) = 2T(n/2) + n n>1

Find an Asymptotic bound on T.

Solution:

We guess the solution is O (n (logn)).Thus for constant 'c'.

$T(n) \leq c\ n\ \log n$

Put this in given Recurrence Equation.

Now,

$T(n) \leq 2c(n/2) \log (n/2) + n$

$\leq cn\log n - cn\log 2 + n$

$= cn\ \log n - n\ (c\log 2 - 1)$

$\leq cn\ \log n$ for $(c \geq 1)$

Thus $T(n) = O\ (n\ \log n)$.